ETERNAL INSTABILITY: STABILITY AS THE LIMIT OF FINITE DWELL TIMES

GTI v2.1 — A Conceptual Framework With One Testable Signature

Author: Nasir Khan
Affiliation: Independent Researcher,Wana,South Waziristan,Pakistan
Contact:03341403026
ORCID: 0009-0002-6006-0796
Version: v2.1 — 12 Oct 2026
Status: No regulator, no stabilizing field, no M_stab. Falsifiable via Josephson ensemble.
Email:nasirk684@gmail.com

ABSTRACT

What is a stable state? We argue that stability is not a property of states at all, but the limit of a

sequence of finite lifetimes. Take a bounded potential containing an infinite ladder of metastable wells. No barrier in such a potential can deepen — yet escape times can still diverge, because the escape saddles widen instead of growing taller. Every level decays at a finite rate $\Gamma_n > 0$; no level achieves $\Gamma = 0$; and the approach to perfect stability becomes a theorem about geometry rather than an assumption about energy scales. The framework commits to exactly one falsifiable prediction: lifetime hierarchies obey $\tau_{n+1}/\tau_n = (1 + 1/n)^{\beta/2}$, or equivalently $\log \tau = -\tfrac{1}{2} \log \kappa$ at fixed barrier height, with the saddle curvature κ independently measurable through the thermal–quantum crossover. The natural testbed is Josephson circuits and flux qubits, and the test can in principle be run on data that already exist. No regulator or stabilizing field appears anywhere in the construction.

1. INTRODUCTION

What is a stable state? The literature offers three answers, and each of them pays a price.

The first is absolute stability: a state is stable if it sits at a global minimum of an energy functional, so that decay is impossible in principle. The trouble is that no fundamental system is known to possess such a minimum. The electroweak vacuum of the Standard Model is metastable for plausible parameter values, and every de Sitter vacuum in semiclassical gravity decays. Absolute stability, it seems, is a property of definitions rather than of anything we have observed.

The second answer is effective stability: a state

counts as stable if its lifetime exceeds the observational window — the age of the universe, or simply the patience of the experimenter. This is pragmatism, not mechanism. The lifetime enters the framework as an input rather than emerging as an output, and the answer quietly concedes that the difference between "stable" and "very long-lived" is a fact about us, not about the system.

The third answer is landscape metastability. In eternal inflation and the string landscape, stability is statistical: an ensemble of vacua with Coleman–de Luccia decay rates, a multiverse of nucleating and decaying regions. Here lifetimes are genuinely computed, but the framework describes a population of systems, and the deepest vacuum of any given sector can still be exactly stable — it may simply have no lighter vacuum to decay into.

This paper proposes a fourth answer, which can be stated in one sentence: stability is the limit of a sequence of finite dwell times in a bounded potential, a limit that diverges because the escape saddles widen. No barrier deepens — in a bounded potential, no barrier can deepen. No regulator, stabilizing field, or supplementary term is introduced; suppression is replaced by waiting. Every level has a decay rate $\Gamma_n > 0$, no level has $\Gamma = 0$, and the approach to perfect stability is a theorem about geometry rather than an assumption about energy scales.

The framework is the survivor of a falsification. Its predecessor, GTI v1, postulated a stabilization mechanism intended to suppress nucleation; the analysis showed that the regulator instead accelerated decay, destroying the very stability it was introduced to protect

(Section 2.4). The correct response to that result is not a better regulator but no regulator at all, and the correct place to look for stability is not among terms added to the potential but in the geometry of the barriers already there.

The argument is deliberately conservative in its machinery. We keep the classical apparatus — overdamped Langevin dynamics, Kramers escape, the exact one-dimensional mean-first-passage formula — and change only the question being asked of it. Section 2 states the postulates and proves that unbounded wandering without stationarity is consistent. Section 3 derives the lifetime hierarchy in closed form: all level-dependence of a dwell time lives in a single saddle integral, scheduling saddle widths gives $\tau_n = \tau_0 W_n$ exactly, and the curvature hierarchy of earlier formulations follows as a derived consequence. Section 4

turns the one falsifiable signature — log τ = −½ log κ at fixed barrier height — into a concrete protocol on Josephson circuits, testable with existing data. Section 5 develops two remarks: black-hole remnants and white holes, where the framework distinguishes "shallow and sharp" from "shallow and flat" for the first time, and the relation to landscape metastability, from which the present framework differs in the one place that matters: here Γ = 0 is unreachable by construction. Section 6 states the falsifiers. Throughout, we hold ourselves to one rule: every claim traces to a postulate, every postulate carries a stated cost, and the framework names in advance the measurement that would kill it.

2. FRAMEWORK

2.1 Postulates

P1 (No true vacuum). V is a bounded, twice continuously differentiable function on the real line. It attains its infimum v_0 at every one of an infinite family of wells a_n, where V(a_n) = v_0 and V''(a_n) = $\omega_0^2 > 0$; the spacing between neighbors is L_n = λW_n with a fixed constant λ ≥ 2. The saddles s_n separating the wells all have the same shape and the same height h = v_0 + ΔV, but a scheduled width: V(s_n + u) = h − ΔV·φ(u/W_n) with W_n = W_0n^{β/2}. Here φ(0) = 0, φ(±1) = 1, and φ''(0) > 0. The minimum is degenerate: it is attained at every level, and no minimum is decay-proof. Escape from every level has a finite rate. Note what is derived rather than assumed — the curvature at the n-th saddle, κ_n = ΔV·φ''(0)/W_n^2, falls as a power law because the widths grow.

P2 (Kick is inevitable). The dynamics is the

overdamped Langevin equation $dX = -V'dt + g\,dW$ with a constant noise strength $g \geq g_0 > 0$. At finite temperature $g^2 = 2k_BT$. At zero temperature the same exponent maps onto the WKB action, $2\Delta V/g^2 \mapsto S_inst/\hbar$ with $S_inst = \int\sqrt{(2m(V-E))}\,dx$. There is no $g \to 0$ limit in nature: something always kicks.

P3 (Stability is a conditional dwell time). The stability of level n means the mean first-passage time from well n to its escape saddle, conditioned on the level. Under P1 and P2 no other stability notion exists, as Section 2.3 shows.

2.2 Construction

The reader should see the object once. Take $\varphi(\xi) = \xi^2 + \xi^4 - \xi^6$ on $|\xi| \leq 1$ and extend it to $\varphi = 1$ for $|\xi| \geq 1$. One checks that $\varphi(0) = 0$, $\varphi(\pm 1) = 1$,

$\varphi''(0) = 2 > 0$, and $\varphi'(\pm 1) = 0$, so the profile meets its plateaus with continuous slope; φ is even and monotone on. At $|\xi| = 1$ the extension has $\varphi' = 0$ but $\varphi'' \neq 0$, so C^2 continuity with the plateau — as at the wells — is supplied by the partition-of-unity glue described next. The wells are harmonic, $V = v_0 + (\omega_0^2/2)(X - a_n)^2$, cut off at a fixed radius ℓ_w; the saddles are the rescaled profile above; and the remaining glue intervals, which are nonempty because $L_n = \lambda W_n$ with fixed $\lambda \geq 2$ and fixed ℓ_w, are filled in with a partition of unity. The result is C^2, bounded between v_0 and h plus a small glue overhead, and all the derivative bounds used below can be read off the construction directly.[0][1]

2.3 Consistency — recurrent, not stationary

Boundedness gives $|V| \leq C$, and the construction gives $|V'| \leq C$ with constants independent of n.

With constant g:

1. Nothing runs away. Scale function $s(x) = \int_0^x e^{2V(y)/g^2}\, dy$ satisfies $s(\pm\infty) = \pm\infty$, because integrand bounded below by $e^{-2C/g^2} > 0$. Recurrent: no explosion, no cascade.

2. No stationary state. Candidate density e^{-2V/g^2} bounded above, cannot be normalized on $\mathbb{R}$. Null recurrent — washboard Brownian motion. Every well visited infinitely often, displacement $\sim\sqrt{t}$, mean recurrence infinite. At large $|X|$ potential $\to$ plateau h, far field free diffusion.

3. Absence of stationarity is a feature. Stationary would require $V \to \infty$, confining

potential P1 forbids. P1+P2 exclude equilibrium; stability only conditional dwell; eternal instability is definition, not failure.

2.4 The v1 autopsy — why there is no regulator

GTI v1 postulated M_stab to multiply nucleation rate by ~10^3 suppression. Kramers $\tau = (2\pi/\sqrt{(\kappa_a \kappa_s)}) \cdot \exp(2\Delta V/g^2)$ has exactly three knobs: barrier height, curvatures, noise strength. Regulator acting on none = fictitious; acting on one = relabeling. Honestly computed, M_stab entered with sign decreasing τ: stabilized nucleated faster, factor 1000 → O(1) wrong sign.

Lesson: any framework obtaining $\Gamma \to 0$ by adding term must answer where term lives among ΔV, κ, g. v2.1 contains no regulator. Suppression replaced by waiting. Survives

honesty by rebuilding from point predecessor fabricated mechanism.

2.5 Dwell time, formally

τ_n = MFPT to escape saddle with reflecting boundary inner side — standard backward-equation. Conditional does epistemic work. With both directions open, inner integral accumulates one equal contribution per well behind start, diverges — null recurrence surfacing inside escape formula. Backflow makes unconditional ill-defined. All stability claims conditional, predictions built from such. Matches experiment: switching ends run. (Appendix A template for backflow.)

3. THE HIERARCHY

3.1 Dwell time as exact first-passage

$T_n(x)$ solves $(g^2/2)T'' - V'T' = -1$, $T(s_n)=0$, reflecting left. Exact:

$$\tau_n = (2/g^2) \int_{a_n}^{s_n} dx\, e^{2V(x)/g^2} \int_{a_n^-}^{x} dy\, e^{-2V(y)/g^2}$$

No approximation, Kramers not assumed.

3.2 Inner integral n-independent

Well harmonic fixed ω_0^2, fixed $v_0 \Rightarrow$ Gaussian n-independent:

$$\tau_n = (2\sqrt{\pi} / g\omega_0)\, e^{-2v_0/g^2} \cdot S_n,\quad S_n \equiv \int_{a_n}^{s_n} e^{2V(x)/g^2}\, dx.$$

All n-dependence lives in saddle integral S_n.

3.3 Saddle integral — width as fundamental

$V(s_n+u)=h-\Delta V\cdot\varphi(u/W_n) \Rightarrow S_n = e^{2h/g^2}$ $W_n\cdot I_\varphi(2\Delta V/g^2)$, $I_\varphi(\lambda)=\int_{-1}^{1} e^{-\lambda\varphi(\xi)}\, d\xi$ dimensionless n-independent. For $\lambda\gg1$ tails exponentially small.

$\tau_n = \tau_0\cdot W_n$, $\tau_0 = (2\sqrt{\pi}/g\omega_0)\cdot e^{2\Delta V/g^2}$ $\cdot I_\varphi(2\Delta V/g^2)$ exact for all n,ΔV,g.

Schedule $W_n=W_0 n^{\beta/2} \Rightarrow \log \tau_n = (\beta/2)\log n$ + const, $\Gamma_n \to 0$, $\Gamma_n>0\ \forall n$.
Curvature derived $\kappa_n=\Delta V\cdot\varphi''(0)/W_n^2=\kappa_0 n^{-\beta}$.
Stability from flatness literal.
Kramers recovered: $I_\varphi\approx\sqrt{(\pi g^2/\Delta V\varphi''(0))} \Rightarrow$ $\tau_n \to (2\pi/\omega_0\sqrt{\kappa_n})\exp(2\Delta V/g^2)$.

3.4 Geometric consistency

Spacing $L_n=\lambda W_n$ $\lambda\geq2$ prevents collision, V

bounded. Curvature-only schedule at fixed regularization saturates as $\kappa_n \to 0$; width is correct fundamental object.

3.5 Ratio law — exact and parameter-free

$\tau_{n+1}/\tau_n = W_{n+1}/W_n = (1+1/n)^{\beta/2}$, no g,T,ħ,φ. At fixed barrier exact. Eliminating index $\kappa_n \propto W_n^{-2}$:

$\log \tau_n = -\frac{1}{2} \log \kappa_n + \text{const}$ — slope −½, nothing fit.

3.6 Quantum regime

$S_n = W_n\sqrt{(2m\Delta V)}\cdot J_\varphi$, $\tau_{q,n} \propto \exp(2\sqrt{(2m\Delta V)}J_\varphi \cdot W_n/\hbar)$ exponential in width — steeper hierarchy. $T_{cr,n}=\hbar\omega_s/2\pi k_B$, $\omega_s=\sqrt{(\kappa_n/m)} \propto n^{-\beta/2}$. Levels below $n_{cr}(T)$ thermal, above quantum, boundary walks down

as T drops. T_{cr,n} measures κ_n directly — second axis of slope test.

3.7 Remarks

If $\Delta V_n \to 0$, correction $I_\varphi(2\Delta V_{n+1}/g^2)/I_\varphi(2\Delta V_n/g^2) \to 1$ asymptotes to fixed-barrier case. Exponential schedule $W_n=W_0 e^{\gamma n/2} \Rightarrow$ ratio $\to e^{\gamma/2}$, quantum $\tau_q \propto \exp(e^{\gamma n/2})$ double hierarchy.

4. TESTBED: JOSEPHSON CIRCUITS

4.1 Single junction, concretely

Phase particle mass $M=(\hbar/2e)^2 C$, $U(\varphi)=-E_J(\cos\varphi + s\varphi)$, $s=I/I_c$. Reference: $I_c=1\mu A$, $C=1pF$, $s=0.95 \to E_J/k_B=23.8K$, $\omega_p/2\pi=4.9GHz$, $\Delta U/k_B=0.50K$, $T_{cr}=37mK$. At

100mK, $\Delta U/k_BT \approx 5$ measurable switching-histogram, $T \approx 2.7T_{cr}$ thermal — standard numbers.

4.2 One junction is one point — and mapping

$X \leftrightarrow \varphi\sqrt{M}$, $\omega_0^2 \leftrightarrow \omega_p^2$, $W \leftrightarrow \Delta\varphi_s \cdot \sqrt{M}$ $\Delta\varphi_s = \sqrt{(2\Delta U/\kappa_\varphi)}$, $\kappa_\varphi = E_J\sqrt{(1-s^2)}$, $g^2 \leftrightarrow 2k_BT$, $\tau_n \leftrightarrow 1/\Gamma(s)$. Along bias $\Delta U \propto (1-s)^{3/2}$, $\omega_p \propto (1-s)^{1/4} \Rightarrow \Delta U \propto \omega_p^6$ cannot tune independently. Hierarchy is ensemble statement, prediction $\log \tau = -\frac{1}{2}\log \kappa + \text{const}$ at fixed ΔU says what ensemble must hold fixed.

4.3 Route (i): ensemble protocol

Protocol A — C-sweep at fixed bias: Hold $I_{c,s}$ fixed (fixed $\Delta U, \kappa_\varphi$), vary C via shunt. $\kappa \propto 1/C$, $W \propto \sqrt{C} \Rightarrow \log \tau = \frac{1}{2}\log C + \text{const}$ slope $+\frac{1}{2}$ vs log C, $-\frac{1}{2}$ vs log κ. $C \in \{0.5,1,2,4\}$pF spans factor 8 in κ,

2.83 in τ measurable. Curvature independent: $T_{cr} \propto C^{-1/2}$ 53 → 19mK from crossover or spectrum. Two spectroscopies, one slope, no fit.

Protocol B — matched-barrier bias leverage: At fixed ΔU $\kappa = \Delta U \cdot r(s)/M$, $r(s) = \sqrt{(1-s^2)}/(2g(s))$, $g(s) = \sqrt{(1-s^2)} - s \cdot \arccos s$, $r \approx 3/(4(1-s))$. r=7.3,14.8,74.8,750 at s=0.9,0.95,0.99,0.999 → two decades κ at fixed barrier via s → 1 with scaled E_J. Protocol A cleaner, B wide-range.

Quantum distinct: log $\tau_q \propto 1/\omega_p \propto \sqrt{C}$ linear in $\kappa^{-1/2}$ not log κ. One ensemble falsifies twice.

Measurement textbook: $P(I_{sw}) \to \Gamma(I)$ inversion standard. Γ(s), $\omega_p(s)$ recorded 40 years → archives re-analyzable without new experiment — cheapest falsification.

4.4 Route (ii): engineered multiwell circuits

Coupled-JJ chains, phase-slip arrays, Kerr oscillators — intrinsic levels, ratio law pointwise. Backflow explicit Appendix A template. Program, not proposal.

4.5 Systematics, and what counts as failure

1. Damping Melnikov–Hänggi $a_t(Q,\Delta U/k_BT)$ dispersion via Q only, Q measurable. −½ leading, a_t systematic not escape hatch — residuals correlating wrong way = failure.
2. Noise window $\Delta U/k_BT$~3-15.
3. Falsification: slope log τ vs log κ at matched $\Delta U \neq -½$ beyond Q-systematic across independent ensembles, or ratio law pointwise failure.

5. REMARKS

5.1 White holes and remnants — with LIGO closure

v1-era claims — white holes forbidden by extra barrier, remnants suppressed $10^3\times$ — died with v1 regulator, nothing in v2.1 revives them. What v2.1 contributes is discipline: persistence is dwell time $\tau=\tau_0 W$ set by geometry of effective well. Shallow and sharp = short τ transient only. Shallow and flat = long τ. Question "are white holes allowed?" is not selection rule but measurement: what are ΔV and saddle geometry?

Framework's honest output given candidate potential is Kramers/MFPT lifetime — never suppression factor vs uncomputed baseline. We

do not derive gravitational effective potential here; any answer "forbidden because extra barrier" must become "τ equals this, given these barrier parameters."

LIGO statement — added per v2.1: Consequently, GTI v2.1 makes no testable prediction for LIGO O4/O5 or upcoming echo searches without an independently derived V_eff for the merger remnant. The white-hole / echo chapter that existed in v1 is closed pending such a derivation. v1's 10^3 suppression claim is formally withdrawn. No stochastic background, echo amplitude, or remnant rate is predicted by v2.1 alone — any future LIGO prediction must first derive ΔV, W, κ from GR + quantum corrections, then apply $\tau=\tau_0 W$.

5.2 Relation to landscape metastability

Closest relative is landscape of eternal inflation, kinship acknowledged: both replace stability with waiting, decay with nucleation. Three differences: (1) lifetime ratios: landscape Coleman–de Luccia actions vs here saddle geometry slope −½ measurable in single system. (2) Γ=0 reachable there deepest vacuum no lighter vacuum, unreachable here P1 forecloses exact stability by construction. (3) Population vs individual: landscape population statistical, v2.1 one system conditional dwell times single trajectory. What landscape gets right v2.1 keeps; what it cannot say — how single system approaches perfect stability and when — answered with saddle width.

6. FALSIFIERS, AND CONCLUSION

F1. Decay-proof state: Γ≡0 as precision

improves, minimum with no escape channel — kills P1.

F2. Hierarchy failure across independent matched-ΔU ensembles — slope log τ vs log κ departing from −½ beyond Q-dispersion — or pointwise ratio law $\tau_{n+1}/\tau_n=(1+1/n)^{\beta/2}$ failure in multiwell device.

Logic two steps forced: P1+P2 exclude stationarity bounded potential no normalizable density ⇒ stability only conditional dwell. Bounded caps barriers ⇒ dwell can diverge only via widening saddles. Everything else — closed-form hierarchy, parameter-free ratio law, slope −½, Josephson protocol — unpacking. If falsified, two negative lessons survive: suppression factors without home among ΔV,κ,g are fabrications, and persistence questions reframed as geometry questions. Framework

that knows what it costs and what survives its death is most conceptual paper can offer. Rest is measurement.

APPENDIX A — MIN-ROUTE BOOKKEEPING

Well n between saddles n−1 (W_{n-1}) and n (W_n), W_n increasing rightward. Escape clock rate $k_m=1/(\tau_0 W_m)$. Dwell competition:

$\tau_n^{dwell} = 1/(k_n+k_{n-1}) = \tau_0 \cdot W_n W_{n-1}/(W_n+W_{n-1})$, reflecting n=1 $k_0=0$ $\tau_1=\tau_0 W_1$ exactly. $W_n=W_0 n^{\beta/2}=W_0 w_n \Rightarrow$

$\tau_{n+1}^{dwell}/\tau_n^{dwell} = (w_{n+1}/w_{n-1})\cdot(w_n+w_{n-1})/(w_{n+1}+w_n) = [1+\beta/n]\cdot[1-\beta/(2n)]+O(n^{-2}) = 1+\beta/(2n)+O(n^{-2})$ identical to single-saddle ratio at leading order.

Escape bias: $W_{n-1}<W_n$ faster clock leftward

⇒ events cascade toward instability. Occupation bias: dwell grows up-ladder ⇒ time dominated by deep levels though events prefer shallow. Walker decays constantly toward instability and perpetually found in stability — thesis in miniature, no extra assumption.

REFERENCES

Kramers 1940; Fulton & Dunkleberger 1974; Coleman 1977; Coleman & de Luccia 1980; Devoret, Martinis et al. 1980s MQT; Linde 1986; Hänggi, Talkner & Borkovec Rev Mod Phys 1990; Bousso & Polchinski 2000; KKLT 2003; Guth 2007.

END GTI v2.1 FINAL